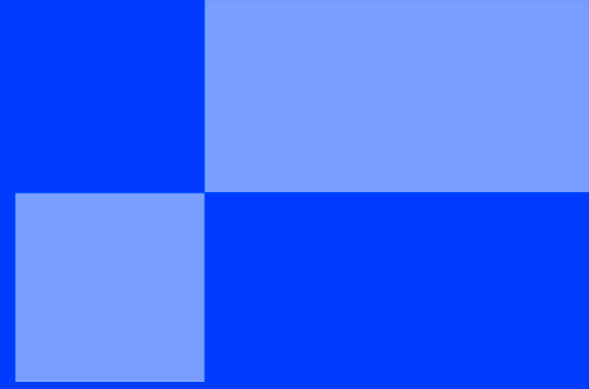
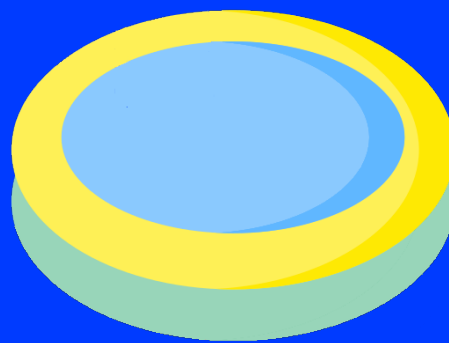




PREPARATION BOOTCAMP

< FIRST PROGRAM />



PREPARATION BOOTCAMP

GAMES



No challenge today, because the entire subject is already a challenge!
That's why you have multiple game proposals.

If you need a little break in your day, we suggest you play:

- ✓ [Python Robocode](#);
- ✓ [robotrubble](#);
- ✓ [lightbot](#);
- ✓ [music lab](#);
- ✓ [leek wars](#).

We also encourage you to resume your game(s) later.



Your objective is to have some fun building (and playing) the guessing game [hangman](#).

Objective and rules of OUR game

Players have to guess a word with a minimum of penalties.

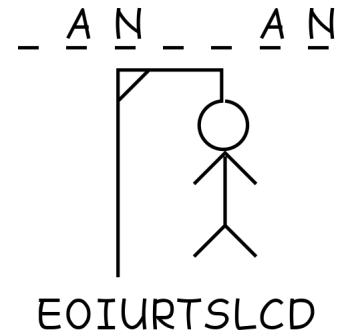
Each turn, a player suggests a letter:

- ✓ if the word contains it, each occurrence of the letter is revealed;
- ✓ if the word does not contain it, the player gets 1 penalty.

At any time, the player can propose a full word:

- ✓ if the word is the one to be guessed, the player wins;
- ✓ else, the player gets 5 penalties.

If the number of penalties exceeds 12, the player loses.



Here is a game example:

```
python3 hangman.py
_ _ _ _ _ / 0 penalty

$> A
Found one 'A'
A _ _ _ _ / 0 penalty

$> e
Found one 'E'
A _ _ _ E / 0 penalty

$> t
No 'T' found
A _ _ _ E / 1 penalty

$> ALIVE
ALIVE: incorrect guess
_ _ _ _ _ / 6 penalties

$> apple
APPLE: correct guess - 6 penalties
```



No graphical interface here, the game must be playable in a terminal.
Feel free to adapt the display of the game as you fancy.

First brick



Write a function that takes an integer as parameter and prints "You lose!" if the parameter is greater than or equal to 12.

Next brick



Write a snippet of code to return a **random** item from the `set {1, 2, 3, 4, 5, 6}`.
Encapsulate this piece of code inside a function.

First package



Install the `english-words-py` package. Then, create a program that:

- ✓ imports the `english_word_lower_set` from this package;
- ✓ prints a random word from this set.



The well-named **PIP** (Package Installer for Python) is the usual tool for this job.

Another brick in the wall

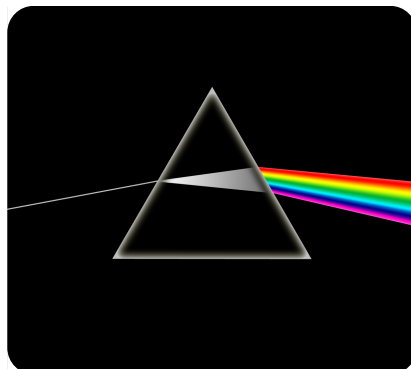


Write a function that:

- ✓ takes a string as parameter;
- ✓ finds the length `n` of this string;
- ✓ prints a string made of `n` pairs `"_ "` (underscore followed by a space).



For instance, the input `PinkFloyd` provides the output `__ _ _ _ _`.



Pseudocode



Before coding a program, a good way to start is to define the **pseudocode**: it will describe the steps your program will follow.

Write down the pseudocode for your hangman game.
Then, test and challenge it with a fellow comrade.



If you're stuck, look for **some inspiration** on the web or in the class.

Implementation



You've defined a few basic functions and described the main steps.
It is now time to dive in. Implement the entire game.
Do as you please to reach this goal.



Never remain stuck.
Ask your fellow comrades or your local pedago or the web.

Give us more



Make your game customizable, by adding a few arguments to its program, such as:

- ✓ the number of penalties
- ✓ the length of the unknown word
- ✓ the theme of the unknown word
- ✓ a file containing a word list to pick in
- ✓ a time limit
- ✓ anything you fancy



A nice tool to parse Python's arguments is **argparse**.

v 2.2

{EPITECH}